

Process Automation System

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1. Problem Statement

Managing files manually in folders such as Downloads or Desktop becomes difficult as the number of files increases. Users often struggle to find files, remove duplicates, create backups, and organize documents efficiently. This project provides an automated file management system that scans folders, categorizes files, removes duplicates, creates backups, generates reports, and maintains logs to improve productivity and reduce manual effort.

2. Objectives

- To automate the file organization process.
 - To reduce manual effort in managing files.
 - To organize files based on their extensions.
 - To detect and remove duplicate files.
 - To automatically rename conflicting files.
 - To generate reports of all file operations.
 - To create automatic backups before modifications.
 - To maintain activity logs.
 - To provide secure login authentication.
 - To develop a simple and user-friendly application.
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3. Features

- User Login Authentication
- Folder Selection
- Automatic File Scanning
- File Type Detection
- Automatic Folder Creation
- Smart File Organization
- Duplicate File Detection
- Automatic File Renaming
- Backup Creation
- CSV Report Generation
- Activity Logging
- File Search
- JSON Configuration

- Menu-Driven System
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4. Module Requirements

Module 1: User Login

Purpose: Allow only authorized users.

Requirements

- User registration and login
 - Password hashing
 - Login validation
 - Store user credentials in **users.json**
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Module 2: Folder Selection

Purpose: Select the folder to organize.

Requirements

- Browse folder path
 - Validate folder exists
 - Store selected folder
 - Display folder information
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Module 3: File Scanner

Purpose: Scan all files.

Requirements

- Read every file inside folder
 - Ignore system files
 - Count total files
 - Prepare file list
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Module 4: File Type Detection

Purpose: Identify every file type.

Requirements

- Detect extension
 - Categorize files
 - Support
 - Images
 - Documents
 - Videos
 - Audio
 - PDF
 - Excel
 - PowerPoint
 - ZIP
 - Programs
 - Others
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Module 5: Automatic Folder Creation

Purpose: Create folders automatically.

Requirements

- Check existing folders
 - Create missing folders
 - Avoid duplicate folder creation
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Module 6: File Organizer

Purpose: Move files.

Requirements

- Move files to correct category
 - Preserve filenames
 - Handle move errors
 - Update report
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Module 7: Duplicate Detection

Purpose: Detect duplicate files.

Requirements

- Calculate SHA-256 hash
 - Compare hashes
 - Report duplicate files
 - Skip or remove duplicates
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Module 8: File Renaming

Purpose: Rename duplicate filenames.

Requirements

- Detect same filename
- Rename automatically
- Maintain naming sequence

Example

```
resume.pdf  
resume_1.pdf  
resume_2.pdf
```

Module 9: Backup Module

Purpose: Backup original files.

Requirements

- Create Backup folder
 - Copy original files
 - Preserve folder structure
 - Restore if needed
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Module 10: Report Generation

Purpose: Generate operation report.

Requirements

- Create CSV report
- Record
- File Name

- Category
 - Status
 - Date & Time
 - Save inside Reports folder
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Module 11: Logging Module

Purpose: Store system activities.

Requirements

- Log every action
 - Store errors
 - Save timestamps
 - Generate log file
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Module 12: Search Module

Purpose: Search organized files.

Requirements

- Search by filename
 - Display file location
 - Fast searching
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Module 13: JSON File Handling

Purpose: Store configuration and user data.

Requirements

- Read JSON files
 - Write JSON files
 - Update JSON data
 - Handle missing files
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Module 14: Main Menu (Integration)

Purpose: Connect all modules.

Requirements

- Display menu
- Call every module
- Handle user choice
- Exit safely

5. Non-Functional Requirements

- User-friendly interface
- Fast execution
- Secure login
- Reliable backup
- Efficient duplicate detection
- Easy maintenance
- Modular programming
- Error handling
- Platform independent
- Scalable architecture

6. Product Backlog Grooming

Product Backlog ID	User Story	Priority	Estimated Effort
PB-01	Login Authentication	High	2 Hrs
PB-02	Folder Selection	High	2 Hrs
PB-03	File Scanner	High	3 Hrs
PB-04	File Type Detection	High	3 Hrs
PB-05	Automatic Folder Creation	High	2 Hrs
PB-06	File Organizer	High	4 Hrs
PB-07	Duplicate Detection	High	3 Hrs
PB-08	File Renaming	Medium	2 Hrs
PB-09	Backup Module	Medium	3 Hrs
PB-10	Report Generation	Medium	2 Hrs
PB-11	Logging Module	Medium	2 Hrs
PB-12	Search Module	Low	2 Hrs
PB-13	JSON Handling	High	2 Hrs
PB-14	Menu Integration	High	3 Hrs
PB-15	Testing & Documentation	High	5 Hrs

7. Sprint Planning

Sprint 1 – Planning Phase (1–2 Days)

- Requirement Analysis
- Project Planning
- Architecture Design
- Module Distribution
- Documentation

Sprint 2 – Development Phase (3–4 Days)

- Login Module
- Folder Selection
- File Scanner
- File Type Detection
- Folder Creation
- File Organizer
- Duplicate Detection
- Renaming Module
- Backup Module
- Report Generation
- Logging Module
- Search Module
- JSON Handling
- Main Menu Integration

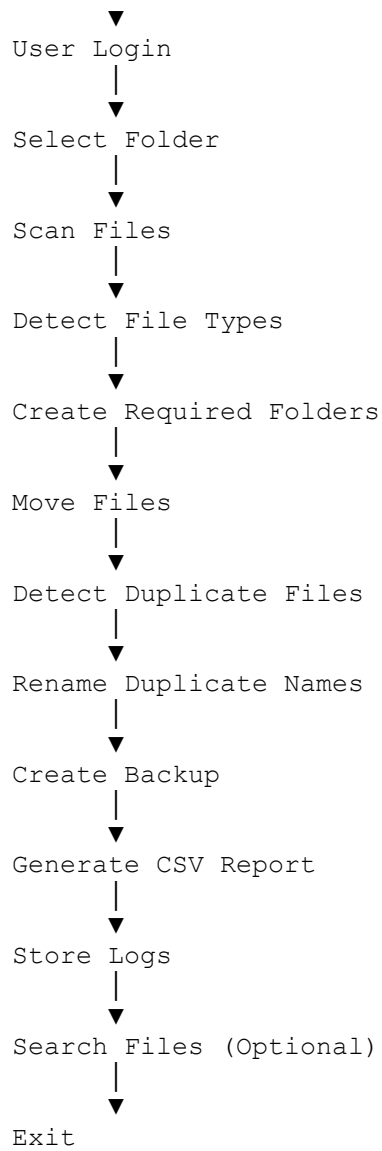
Sprint 3 – Testing Phase (1–2 Days)

- Functional Testing
- Bug Fixing
- Integration Testing
- Validation
- Final Demonstration

8. Workflow

Start

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9. Technologies Used

- Python
 - JSON
 - os
 - shutil
 - pathlib
 - hashlib
 - logging
 - csv
 - datetime
 - VS Code
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10. Acceptance Criteria

- User login works correctly.
 - Folder is selected successfully.
 - Files are scanned accurately.
 - Files are categorized correctly.
 - Required folders are created automatically.
 - Files are moved without data loss.
 - Duplicate files are detected accurately.
 - Duplicate filenames are renamed correctly.
 - Backup is created successfully.
 - CSV report contains correct details.
 - Activity log records all operations.
 - Search returns the correct file location.
 - The system handles invalid inputs without crashing.
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11. Future Scope

- AI-based file categorization
- Automatic cloud backup (Google Drive/OneDrive)
- File compression before backup
- Real-time folder monitoring
- Drag-and-drop interface
- File preview feature
- Password-protected backups
- Duplicate image detection using AI
- Dashboard with charts and statistics
- Email notification after completion